PHARMAWATCH: MULTI-MODAL PHARMACOVIGILANCE FINGERPRINTING & DYNAMIC VIGILANCE INDEX (DVI)

Complete Mathematical Formulation, Variable Definitions, Evidentiary Authenticity Architecture, and Comparative Methodological Rationale

Version: 1.0.0 (Approved Scientific Specification)	**Target System:** PharmaWatch Real-Time Engine
Classification: FDA 21 CFR Part 11 Compliant Architecture	**Methodology:** Empirical Bayes + Multi-Modal Meta-Fusion

1. Executive Summary & Foundational Objective

Modern post-marketing pharmacovigilance (PV) is paralyzed by two opposing structural failures: **Alert Fatigue** and **Delayed Crisis Detection**. Classical disproportionality methods such as raw Proportional Reporting Ratio (PRR) flag thousands of statistically noisy or trivial signals each quarter, while genuine emergent toxicities remain buried in backlogs.

The foundational objective of PharmaWatch is to deploy an authentic, mathematically sound Multi-Modal Fingerprinting framework that synthesizes spontaneous reporting (FAERS), clinical trial registries (ClinicalTrials.gov), peer-reviewed literature (PubMed), and early community discussions (Reddit) into a unified, actionable **Dynamic Vigilance Index (DVI)**.

> **[THE TWO-LAYER TRUTH PROTOCOL]**
> PharmaWatch enforces strict evidentiary isolation: Layer 1 maintains 100% pristine, unblended, auditable regulatory ground truth (exact 2x2 counts, ROR, PRR, PMIDs, NCT IDs). Layer 2 operates strictly as an operational triage filter (DVI and 4D Fingerprint) to prioritize clinical review without data pollution.

2. Evidentiary Authenticity & Provenance Framework

In clinical pharmacology, averaging unverified consumer forum chatter directly with double-blind randomized clinical trial results invalidates the scientific validity of the dataset. Regulators require strict data provenance.

PharmaWatch avoids data pollution through **Epistemic Tiering**. High-rigor experimental trials receive dominant causality weights, while social media is quarantined as an Early Sentinel Smoke Detector. Every composite calculation provides direct clickable traceability back to primary repository IDs. This aligns with the **FDA Sentinel Initiative**, the **European Commission Web-RADR Project**, and **CIOMS Working Group VIII** guidelines.

3. The 4 Universal Dimensions of the Safety Fingerprint [S / V / H / C]

Every drug-event association is characterized across four standardized dimensions, each bounded between 0.0 and 25.0 points:

3.1 Dimension 1: Statistical Strength (S_stats)

Calculated using the conservative lower bound of the 95% Confidence Interval of the Reporting Odds Ratio (ROR_lower_95) to automatically penalize small sample sizes:

```
ROR = (a * d) / (b * c)
SE(ln_ROR) = sqrt( 1/a + 1/b + 1/c + 1/d )
ROR_lower_95 = exp( ln(ROR) - 1.96 * SE(ln_ROR) )

If ROR_lower_95 <= 1.0 or a < 3:
S_stats = 0.0
Else:
S_stats = min( 25.0, 8.0 * ln(ROR_lower_95) )
```

Methodological Rationale: Raw PRR explodes to infinity when report numbers are tiny (e.g., a=2, b=0). By utilizing ROR_lower_95, the massive standard error associated with small counts automatically drives the lower bound below 1.0, setting S_stats = 0.0 and preventing false-alarm cascades. Chi-Square alone was rejected because in 25M-record databases, clinically trivial differences produce p < 10^-15 purely due to sample size.

3.2 Dimension 2: Signal Velocity & Kinematic Momentum (V_velocity)

Evaluates the first and second derivatives of reporting frequency over moving 6-month windows to distinguish static historical signals from accelerating outbreaks:

```
v = (N_recent - N_prior) / max(1, N_prior) # Velocity (Growth Rate)
alpha = (N_recent - 2 * N_prior + N_base) / max(1, N_base) # Acceleration (Momentum)
K_raw = 0.65 * v + 0.35 * alpha

If K_raw <= 0:
V_velocity = 2.0 (Stable historical baseline)
Else:
V_velocity = min( 25.0, 5.0 + 10.0 * ln(1.0 + K_raw) )
```

Methodological Rationale: Real-world safety disasters follow non-linear sigmoidal curves. Factoring both velocity and acceleration detects the critical inflection point of an epidemic months before raw cumulative report totals catch up. Simple Moving Averages were rejected as lagging indicators.

3.3 Dimension 3: Clinical Harm & Outcome Severity (H_harm)

Weights standardized MedDRA outcome codes (OUTC_COD) according to medical danger:

```
Harm_Numerator = (1.00*Count_DE) + (0.85*Count_LT) + (0.60*Count_HO) +
(0.50*Count_DS) + (0.70*Count_CA) + (0.40*Count_RI) + (0.10*Count_OT)

Harm_Ratio = Harm_Numerator / max(1, Total_Evaluated_Outcomes)
H_harm = min( 25.0, 25.0 * Harm_Ratio )
```

Outcome Weights: DE (Death, 1.00), LT (Life-Threatening, 0.85), HO (Hospitalization, 0.60), DS (Disability, 0.50), CA (Congenital Anomaly, 0.70), RI (Required Intervention, 0.40), OT (Other, 0.10). Continuous harm weighting ensures high-mortality events receive top triage priority.

3.4 Dimension 4: Causal Biological Plausibility (C_causality)

Integrates clinical dechallenge/rechallenge, peer-reviewed literature, and molecular graph interaction models (Bradford Hill criteria):

```
Score_Dechal = 6.0 * P_dechal + 4.0 * P_rechal (10.0 pts max)
Score_Lit = min( 8.0, 2.5 * ln( 1.0 + N_pubmed ) ) (8.0 pts max)
Score_GNN = 7.0 * GNN_conf (7.0 pts max)

C_causality = min( 25.0, Score_Dechal + Score_Lit + Score_GNN )
```

4. Source-Specific Mapping & Harmonization Adapters

Source	Harmonization Formula into [S, V, H, C]

openFDA FAERS	`S = min(25, 8*ln(ROR_lower))`
	`V = min(25, 5 + 10*ln(1 + K_raw))`
	`H = 25 * Harm_Ratio (DE, LT, HO outcomes)`
	`C = 6*P_dechal + 4*P_rechal`

ClinicalTrials.gov	`S = min(25, (Drug_Rate - Placebo_Rate) * 250)`
	`V = Recency (25 pts if <6mo; 5 pts if >2yr)`
	`H = 25 * (Serious Adverse Events / Total AEs)`
	`C = Trial Design Rigor (Phase 3 RCT = 25; Open-label = 10)`

PubMed (BioBERT)	`S = min(25, 5*ln(1+Mentions) + 10*Density)`
	`V = Publication momentum (last 2yr vs 5yr base)`
	`H = BioBERT severe entity tags (mortality, necrosis)`
	`C = Oxford CEBM Hierarchy (Meta-analysis = 25; Case report = 6)`

Reddit (PRAW)	`S = min(25, 4*ln(1+Patient_Mentions))`
	`V = 30-day post surge vs 6-month baseline`
	`H = NLP crisis language density ('ER', 'hospital')`
	`C = Self-reported dechallenge (Strictly capped at 8.0 max)`

5. Cross-Source Meta-Fusion & Consensus Mathematics

Each stream is weighted by its scientific reliability: **W_trials = 0.35**, **W_faers = 0.30**, **W_pubmed = 0.25**, **W_reddit = 0.10**.

```
Master_S = (0.35 * S_trials) + (0.30 * S_faers) + (0.25 * S_pubmed) + (0.10 * S_reddit)
Master_V = (0.35 * V_trials) + (0.30 * V_faers) + (0.25 * V_pubmed) + (0.10 * V_reddit)
Master_H = (0.35 * H_trials) + (0.30 * H_faers) + (0.25 * H_pubmed) + (0.10 * H_reddit)
Master_C = (0.35 * C_trials) + (0.30 * C_faers) + (0.25 * C_pubmed) + (0.10 * C_reddit)

If Master_S == 0.0:
DVI_Master = min( 15.0, (Master_H + Master_C) / 2.0 )
Else:
DVI_Master = Master_S + Master_V + Master_H + Master_C
```

Discrepancy Analysis (Hidden Insights Engine):
* **Pattern 1 (Early Sentinel Spike):** Reddit_V > 20.0 and FAERS_S < 5.0 indicates an emerging patient outbreak before clinical reporting.
* **Pattern 2 (Litigation / Media Bias):** FAERS_S > 20.0 and Trials_C < 5.0 indicates news or legal advertising driving artificial reporting.
* **Pattern 3 (Boxed Warning Hazard):** All sources High Harm and High Causality confirms true clinical toxicity.

6. Target Audience Segmentation & Operational Deployment

1. Primary Commercial Target: Mid-sized pharmaceutical and biotech firms (50 - 1,000 employees) that cannot justify $150k+/year for Oracle Argus or ArisGlobal, but require real-time streaming surveillance to protect their post-marketing pipeline.
2. Secondary Commercial Audience: Contract Research Organizations (CROs) monitoring dozens of post-approval clinical protocols simultaneously.
3. Clinical Practice: Hospital pharmacologists checking multi-drug regimens using the GNN graph predictor.
4. Academic Community: Pharmacoepidemiologists publishing reproducible disproportionality studies.

7. End-to-End Numerical Case Studies

Case Study A: Montelukast & Neuropsychiatric Events (Suicidal Ideation)
FAERS: [21.4 / 18.0 / 16.5 / 5.7] | Trials: [22.5 / 8.0 / 4.5 / 25.0] | PubMed: [18.8 / 14.0 / 20.0 / 22.0] | Reddit: [20.0 / 22.5 / 18.0 / 6.5]
Master Consensus: Master_S = 21.0 | Master_V = 14.0 | Master_H = 13.3 | Master_C = 16.6
FINAL DVI MASTER SCORE = 64.9 / 100 [ACTIVE INVESTIGATION / REGULATORY ALERT]

Case Study B: Metformin & Transient Dyspepsia / Nausea
Master_S = 24.5 | Master_V = 2.0 | Master_H = 2.5 | Master_C = 12.0
FINAL DVI MASTER SCORE = 41.0 / 100 [WATCHLIST / BENIGN BASELINE]
Clinical Takeaway: Raw PRR triggers a false-positive emergency alarm on Metformin (PRR > 15.0). The DVI fingerprint correctly categorizes it as low-velocity, non-severe tolerability noise.

8. Master Variable Dictionary & Scientific Boundaries

Variable	Dimension	Range	Definition
`a, b, c, d`	S (Stats)	[0, Inf)	2x2 contingency counts from FAERS or clinical registries
`ROR_lower_95`	S (Stats)	[0.0, Inf)	95% CI lower bound of Reporting Odds Ratio
`v`	V (Velocity)	(-1.0, Inf)	First-order reporting velocity (rate of growth)
`alpha`	V (Velocity)	(-Inf, Inf)	Second-order kinematic acceleration / momentum
`Harm_Ratio`	H (Harm)	[0.10, 1.00]	MedDRA outcome-weighted clinical severity index
`P_dechal`	C (Causality)	[0.0, 1.0]	Proportion of cases with positive dechallenge
`GNN_conf`	C (Causality)	[0.0, 1.0]	Molecular interaction probability from PyTorch GNN
`DVI_Master`	Master Index	[0.0, 100.0]	Composite Dynamic Vigilance Index (0-100)